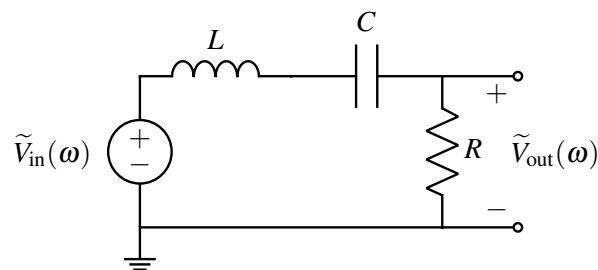

EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 6B

For this discussion, **Note 5** and **Note 6** are helpful.

1. Band-pass filter

It is quite common to need to design a filter which selects only a narrow range of frequencies. One example is in WiFi radios, it is desirable to select only the 2.4GHz frequency containing your data, and reject information from other nearby cellular or bluetooth frequencies. This type of filter is called a band-pass filter; we will explore the design of this type of filter in this problem.



- (a) **Write down the impedance of the series RLC combination in the form $Z_{RLC}(\omega) = A(\omega) + jX(\omega)$, where $A(\omega)$ and $X(\omega)$ are real valued functions of ω .**

- (b) **Write down the transfer function $H(\omega) = \frac{\tilde{V}_{out}(\omega)}{\tilde{V}_{in}(\omega)}$ for this circuit.**

- (c) **At what frequency ω_n does $X(\omega_n) = 0$?** (i.e. at what frequency is the impedance of the series combination of RLC purely real — meaning that the imaginary terms coming from the capacitor and inductor completely cancel each other. This is called the *natural frequency*.)

What happens to the relative magnitude of the impedances of the capacitor and inductor as ω moves above and below ω_n ? What is the value of the transfer function at this frequency ω_n ?

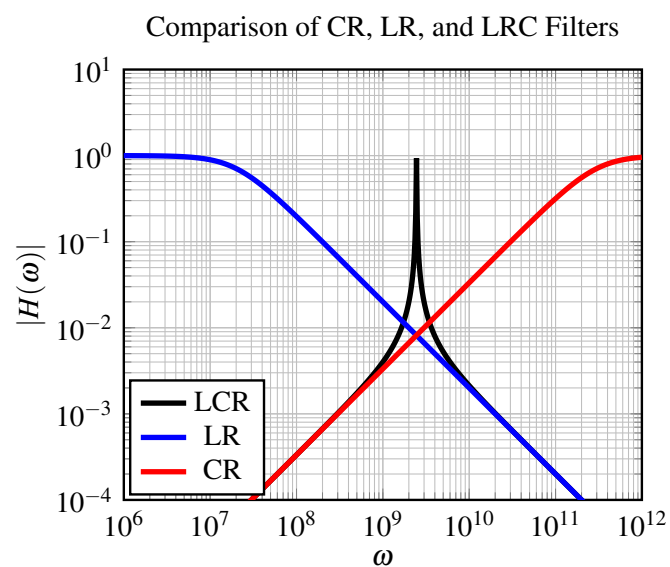
- (d) In most filters, we are interested in the cutoff frequency, since that helps define the frequency range over which the filter operates. Remember that this is the frequency at which the magnitude of the transfer function drops by a factor of $\sqrt{2}$ from its maximum value. Recalling that $H(\omega)$ can be written in the form $H(\omega) = \frac{R}{R + jX(\omega)}$, so that $|H(\omega)| = \frac{R}{\sqrt{R^2 + X(\omega)^2}}$, we see that $|H(\omega)| \leq 1$. Thus, the cutoff frequency is ω_c such that $|H(\omega_c)| = \frac{1}{\sqrt{2}}$, which is when $X(\omega_c) = \pm R$.

We want to find such ω_c . To do this, we want to see what happens in the neighborhood of ω_n and so express the combined effect of the capacitor and inductor in terms of $\omega = \omega_n + \Delta\omega$, where $\Delta\omega$ is (presumably) a relatively small number compared to ω_n .

Write an expression for $X(\omega_n + \Delta\omega)$, where $\Delta\omega$ is a variable shift from ω_n . Find the values of $\Delta\omega_1$ and $\Delta\omega_2$ which give $X(\omega_n + \Delta\omega_1) = -R$, and $X(\omega_n + \Delta\omega_2) = +R$. You may use the approximation that $\frac{1}{1+x} \approx 1 - x$ if $x \ll 1$.

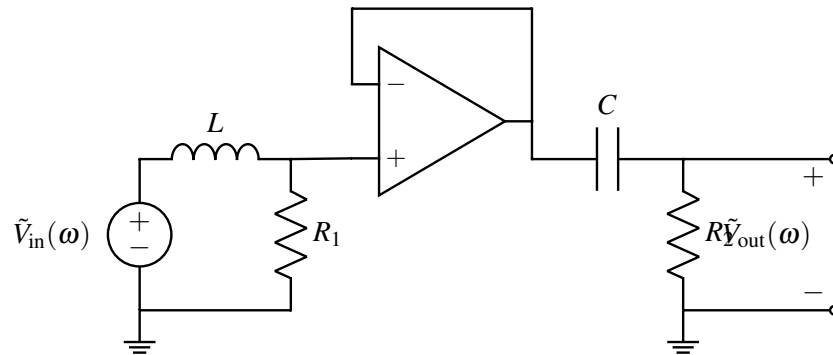
- (e) **Simplify $X(\omega)$ in two cases, when $\omega \rightarrow \infty$ and when $\omega \rightarrow 0$. Plug this simplified $X(\omega)$ into your previously solved expressions to find the transfer function at high and low frequencies.**

Our band-pass filter looks like an LR low-pass filter at high frequencies and a CR high-pass filter at low frequencies. Note that in this case, the cutoff frequencies for the LR and CR filters are not the same as LCR cutoff frequencies or the resonance frequency. In the vicinity of the resonance frequency, notice that the filter is much sharper than any first-order LR or RC filter.



2. Bode Plots

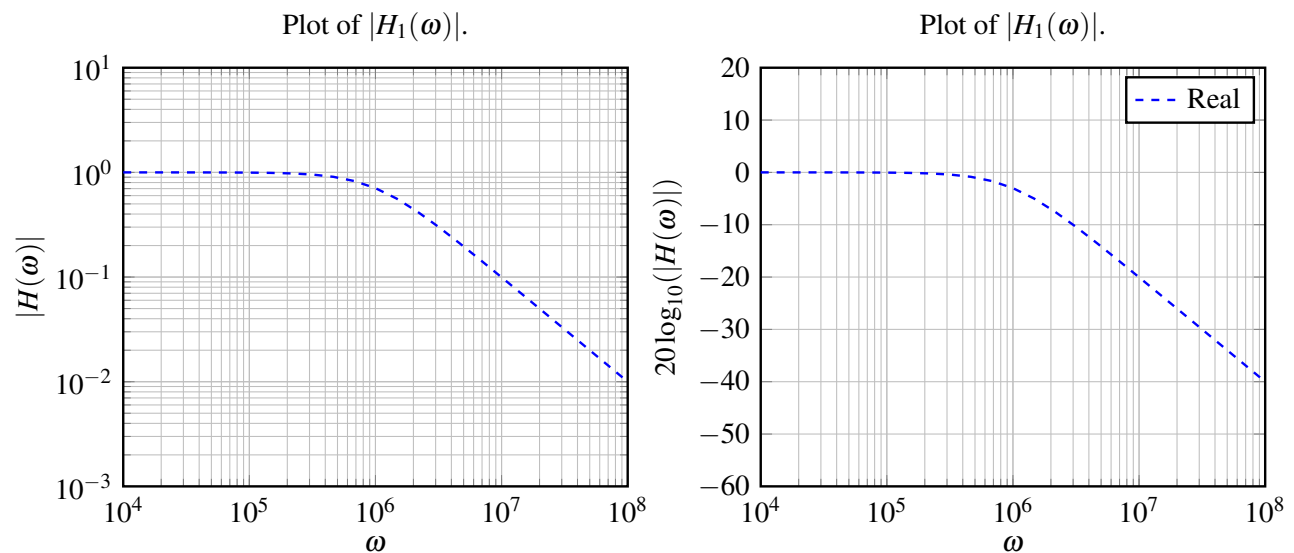
Our eventual goal is to construct Bode plots of the following circuit, with $L = 100\mu\text{H}$, $C = 1\mu\text{F}$, $R_1 = 100\Omega$, and $R_2 = 1\text{k}\Omega$:



To do this we will leverage the fact that Bode plots can be composed in simple ways.

Before we start diving into the problem, let's consider a modification of the *magnitude* plot that will help us work with multiple magnitude plots at once. Namely, instead of plotting $|H(\omega)|$ vs. ω where the y-axis is on a *logarithmic* scale, we plot $20\log_{10}(|H(\omega)|)$ vs. ω instead, and now the y-axis is on a *linear* scale.

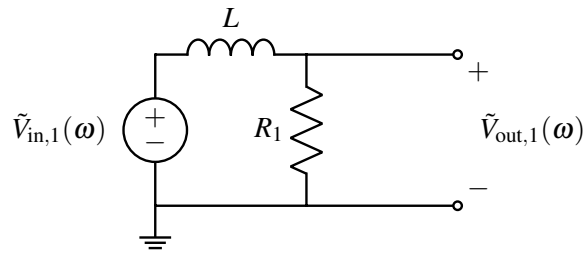
The reason that we do this is that, when combining transfer functions, we end up multiplying them. But we really want to add two plots graphically, not multiply them, so we will just plot and add the logarithms. (The constant multiple 20 is there for convention reasons.) Here's what this looks like, with the old grid on the left, and the new grid on the right:



Notice that we do not need to do this for the *phase* plots, since their y axes are naturally in linear scale, and combining plots can already be done by addition.

Now we are ready to begin working on the problems.

- (a) Consider the first half of this circuit:



We learned in the previous discussion that the transfer function is given by

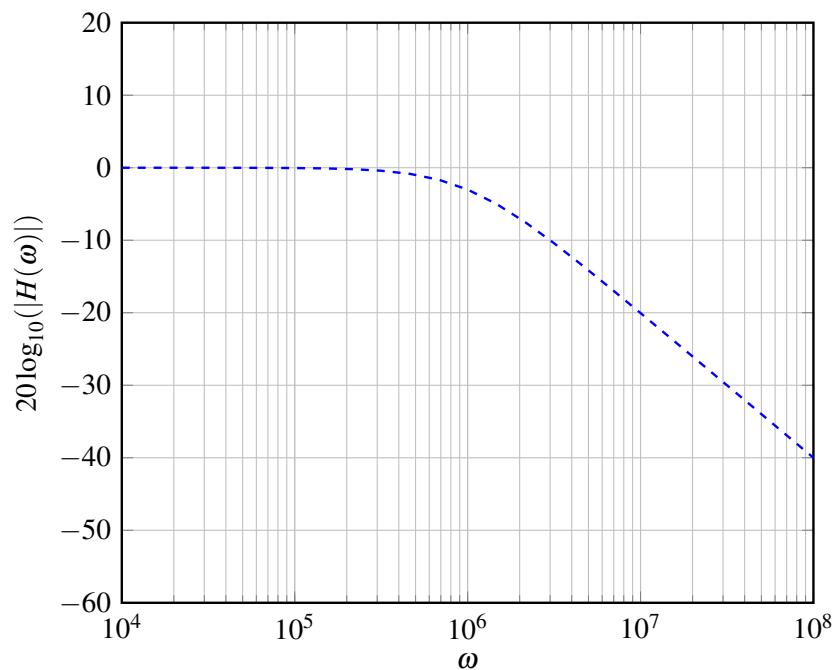
$$H_1(\omega) = \frac{\tilde{V}_{out,1}}{\tilde{V}_{in,1}} = \frac{1}{1 + j\omega \frac{L}{R_1}},$$

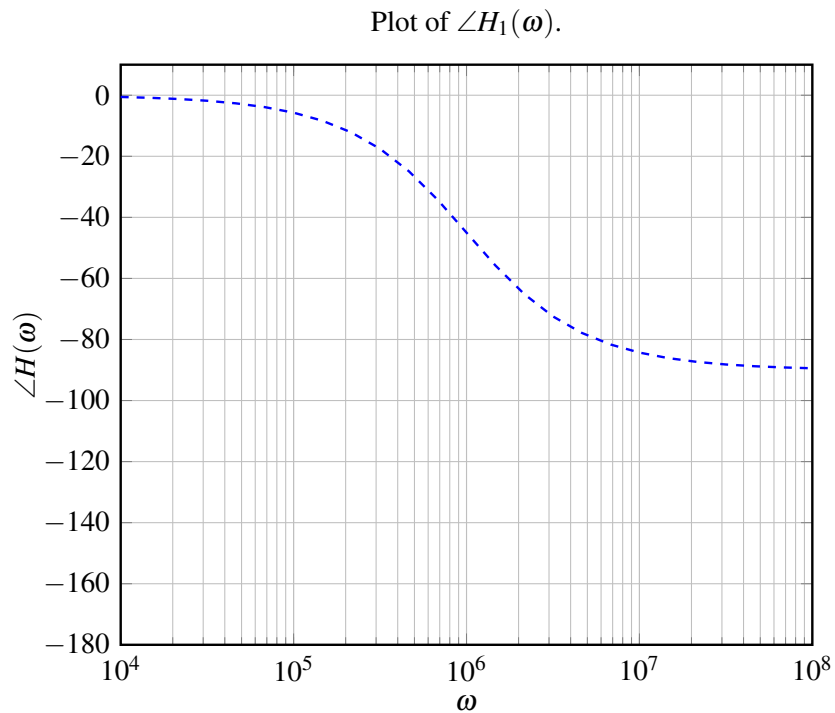
the cutoff frequency $\omega_{c,1}$ is given by

$$\omega_{c,1} = \frac{R_1}{L} = \frac{100\Omega}{100\mu\text{H}} = 1 \times 10^6 \frac{\text{rad}}{\text{s}},$$

and plots of the transfer function are given by

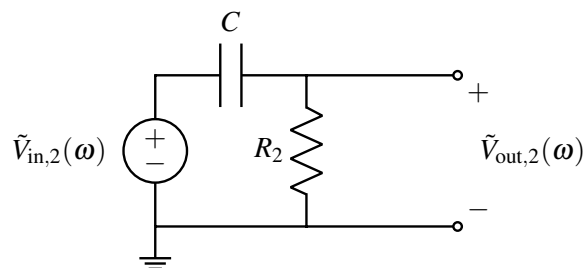
Plot of $|H_1(\omega)|$.





On these grids, **draw the Bode plots for magnitude and phase.**

(b) Consider the second half of the circuit:



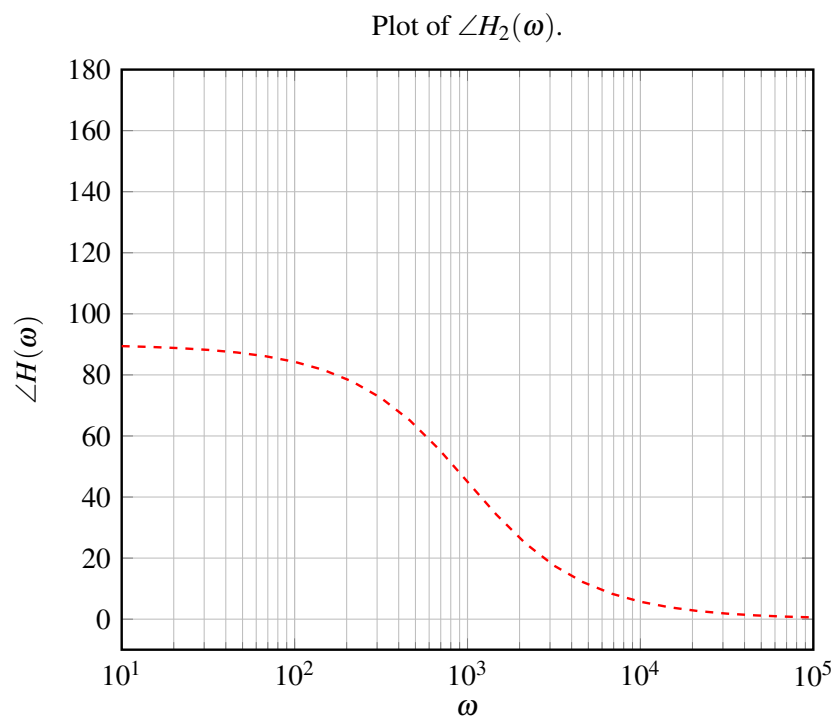
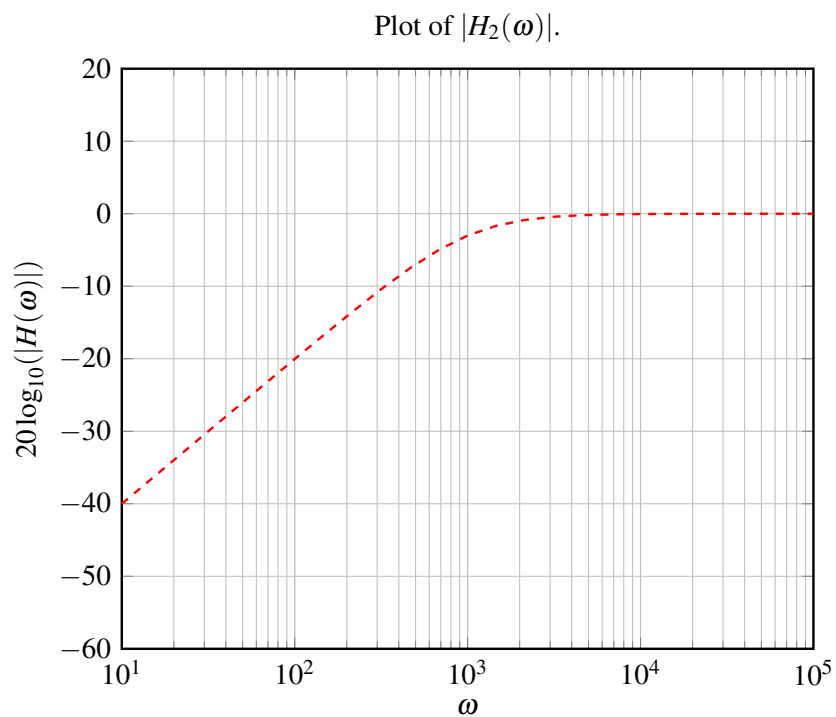
We learned in the previous discussion that the transfer function is given by

$$H_2(\omega) = \frac{\tilde{V}_{\text{out},2}}{\tilde{V}_{\text{in},2}} = \frac{j\omega R_2 C}{1 + j\omega R_2 C},$$

the cutoff frequency $\omega_{c,2}$ is given by

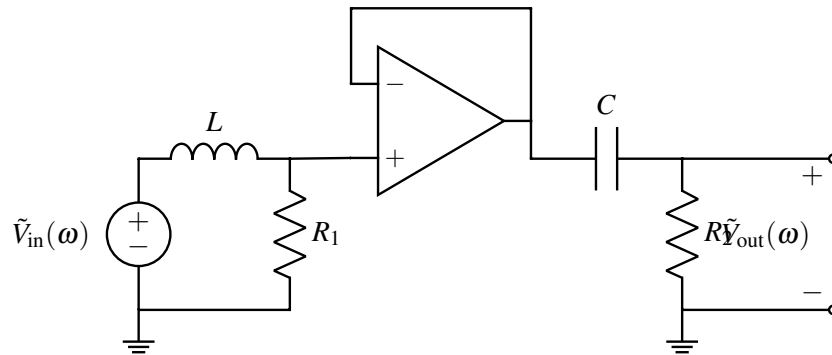
$$\omega_{c,2} = \frac{1}{R_2 C} = \frac{1}{(1\text{ k}\Omega) \cdot (1\text{ }\mu\text{F})} = 1 \times 10^3 \frac{\text{rad}}{\text{s}},$$

and plots of the transfer function are given by



On these grids, **draw the Bode plots for magnitude and phase.**

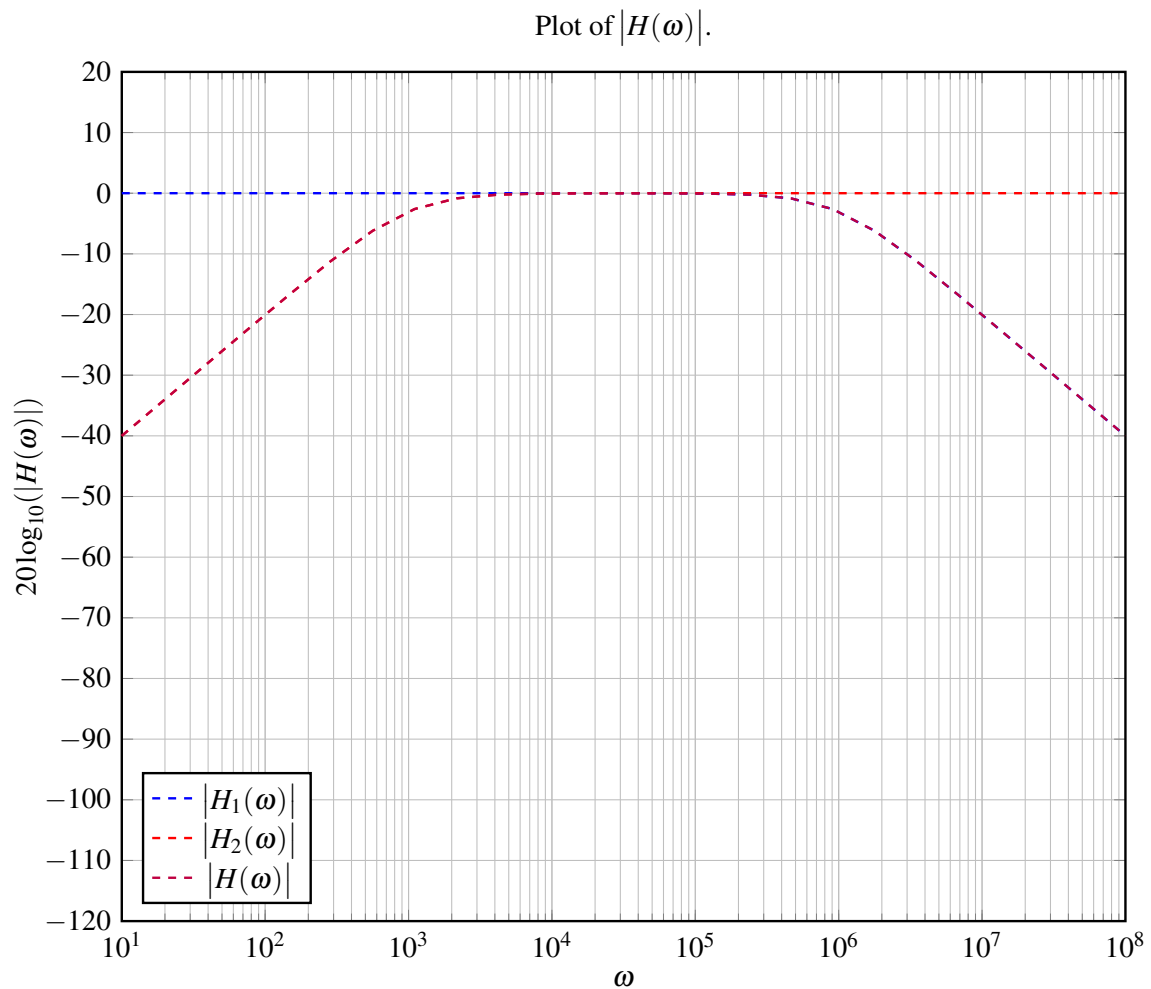
(c) Now, we will put this circuit together. Recall the original diagram:

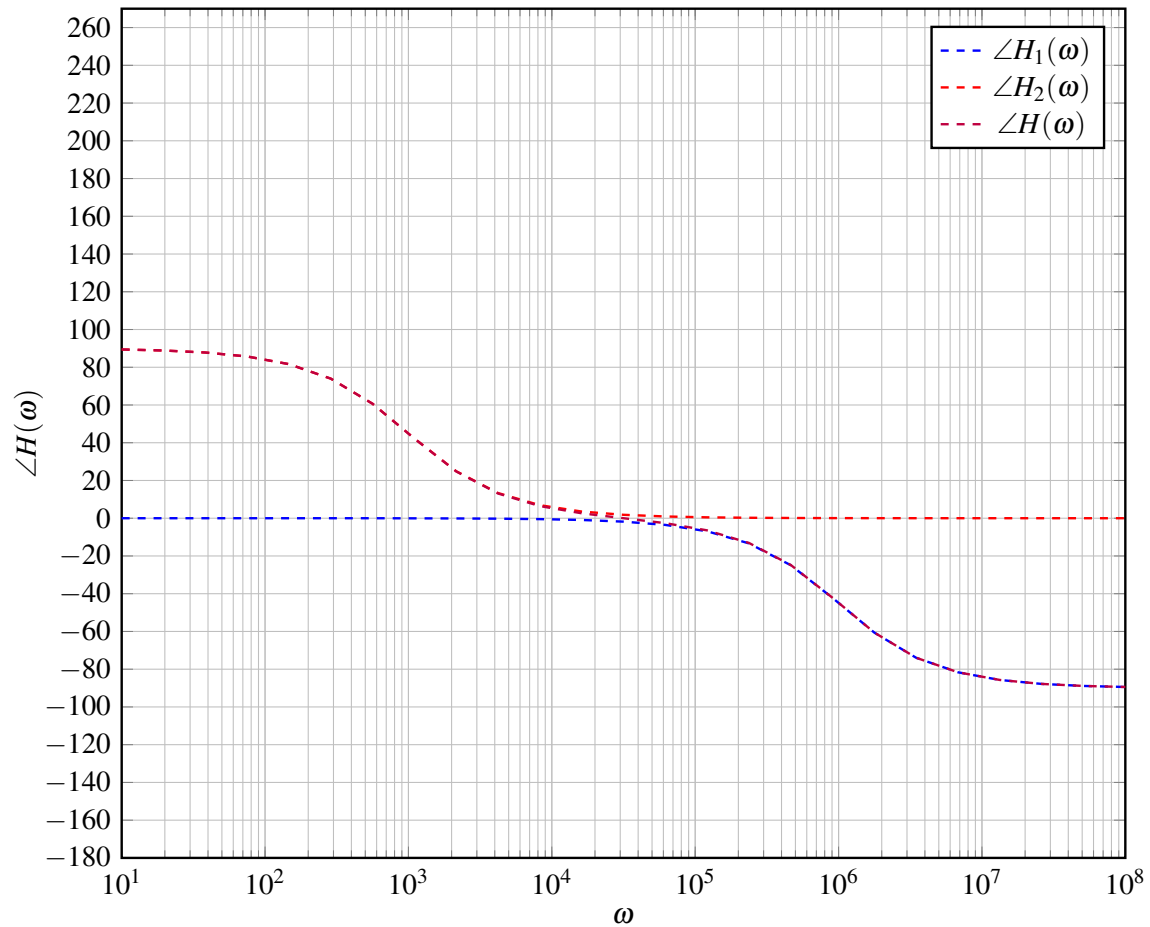


We learned in the previous discussion that the transfer function is

$$H(\omega) = \frac{\tilde{V}_{out}}{\tilde{V}_{in}} = H_1(\omega)H_2(\omega)$$

and the transfer function plots are given by



Plot of $\angle H(\omega)$.

On these grids, **draw the Bode plots for magnitude and phase.**

Hint: Recall that

$$\begin{aligned}
 20\log_{10}(|H(\omega)|) &= 20\log_{10}(|H_1(\omega)H_2(\omega)|) = 20\log_{10}(|H_1(\omega)| |H_2(\omega)|) \\
 &= 20\log_{10}(|H_1(\omega)|) + 20\log_{10}(|H_2(\omega)|) \\
 \text{and } \angle H(\omega) &= \angle H_1(\omega) + \angle H_2(\omega).
 \end{aligned}$$

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